

Does a “Fuzzy Soil Index” Help a Robot Walk on Soft Ground?

A replication study of the *wcci2026_hunter* fuzzy soil model

Hunter humanoid / IsaacSim / HumanoidVerse

June 15, 2026

The problem (from the paper)

Goal: make a humanoid robot walk on **agricultural soil** — furrowed, deformable ground where the feet sink and slip.

- Real soil is **not uniform**: it changes from firm & grippy to soft & slippery.
- Two things vary independently:
 - **Traction** — how grippy the surface is (friction μ).
 - **Support** — how firm it is / how much the foot sinks (stiffness).
- The robot must **judge how hard the current soil is** to adapt and stay upright.

The paper's pitch

Combine traction + support into one human-readable “**soil difficulty score**” using **fuzzy logic**, and use it to drive training / reduce foot slip (>95% claimed).

What the “fuzzy soil index” actually is

A **fuzzy (Mamdani) controller**: two inputs $\rightarrow$ one difficulty score.

- Inputs: **traction** μ and **support** (stiffness).
- Each is graded *Low / Med / High* with **smooth** (fuzzy) boundaries.
- Rules $\rightarrow$ soil is *Easy / Moderate / Hard*.
- Output: one number $d \in [0.2, 0.8]$ (higher = harder).

Selling points: interpretable bins, smooth interpolation, blends two factors.

Two soil “dials”

traction μ & support (firmness)

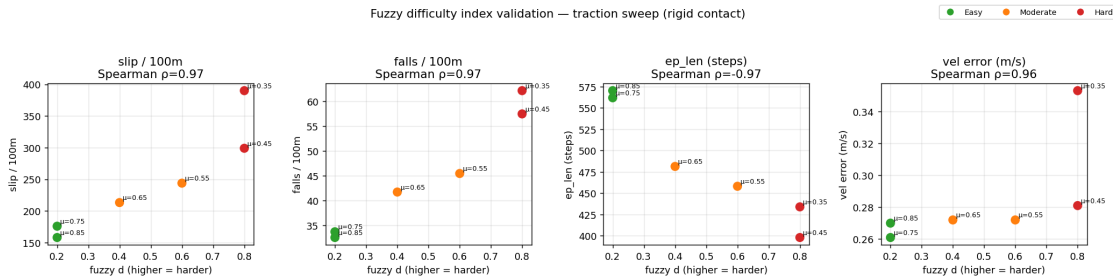
$\downarrow$ fuzzy rules $\downarrow$

difficulty = Easy / Mod / Hard

First impression: the fuzzy index looks convincing

Sweep the soil from grippy to slippery, and plot the fuzzy score d against the **real measured difficulty**:

Fuzzy difficulty index validation — traction sweep (rigid contact)



d tracks difficulty almost perfectly (slip, falls, episode length, velocity error; Spearman rank correlation $|\rho| \approx 0.97$ — same ordering of soils). **This is the chart that makes the fuzzy index look good** — but it only shows d tracks *one* axis; whether that beats the raw numbers is the next question.

Does the fuzzy index **actually add value** over just using the **raw soil numbers**?

- A fuzzy score is only useful if it **beats the raw measurements** it is built from.
- We replicate the recipe on the **Hunter** humanoid in **IsaacSim** (HumanoidVerse) and test the index four ways.
- We judge it as a **difficulty descriptor** and as a **control signal**.

Honest-negative friendly: we want the right answer, not a flattering one.

How we tested it — five settings

Study	Simulator	Fuzzy used as	Support axis
A	rigid PhysX	passive label	physically inert
B	rigid PhysX	curriculum controller	inert
C	DFH deformable	descriptor	physically active
① ②	DFH (+ noisy sensing)	descriptor / decisions	active
D	DFH	redesigned (fixed) index	active

Each study closes the previous one's loophole. "Support axis" = whether soil *firmness* actually does anything in that simulator. ①② = noisy-sensing probes on Set C's soil; **D** = we *fix* the broken rules and re-test.

Sets A & B (rigid simulator)

Set A — fuzzy as a label: does d track difficulty / beat raw μ ?

- d orders soil correctly (Spearman $\rho \approx 0.97$) ...
- ... but adds **nothing** over raw friction μ .
- **Why:** rigid PhysX has **no real “firmness” axis** — the foot can't sink, so the 2-dial model collapses to 1 dial (friction).

Set B — fuzzy drives the training curriculum:

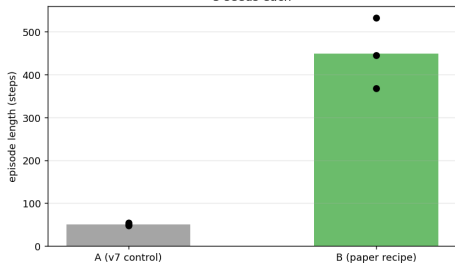
- fuzzy pacing $\approx$ crisp threshold $\approx$ fixed schedule (no significant difference).

But the $\rho \approx 0.97$ chart looked so good — what's going on? →

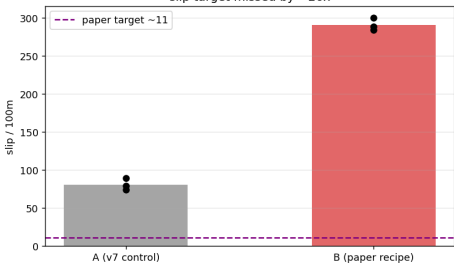
Set A/B — the full evidence (rigid simulator)

Fuzzy-Soil-on-Furrows — experiment summary (all data from source CSVs)

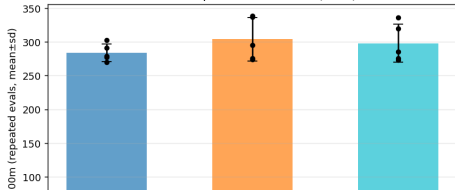
(1) Survival REPLICATES — B walks 8.8x longer
3 seeds each



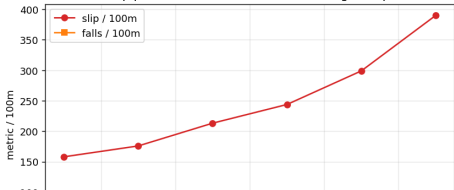
(2) Slip does NOT replicate — B is WORSE, not <11
slip target missed by ~26x



(3) Gate INERT, compliance WASH
all overlap within eval noise (~8%)



(4) Fuzzy difficulty tracks FRICTION
slip $\rho=+0.97$ vs d — but d adds nothing over μ



The catch behind the pretty chart

The $\rho \approx 0.97$ looks great — but it is **only the traction (friction) axis**.

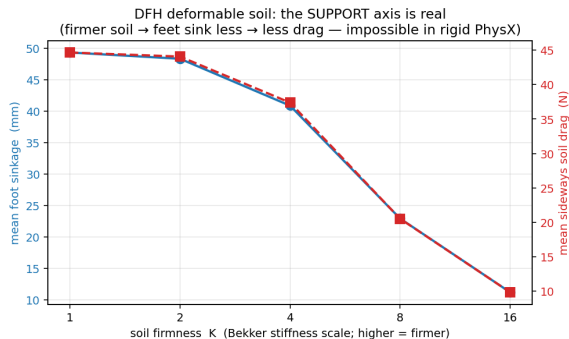
- In rigid PhysX the **support/firmness axis does nothing** (feet can't sink).
- So d is really just a **re-labelling of friction** μ — it carries **no information that μ doesn't already give**.
- A 2-input fuzzy model can't earn its keep when **one input is dead**.

So we need...

a simulator where soil **firmness is physically real** — next.

The fix: DFH deformable soil

DFH (Deformable Furrowed Heightfield) adds the missing physics: feet **sink** (Bekker model: compaction under the foot), generate **sideways drag**, and the soil **deforms**.



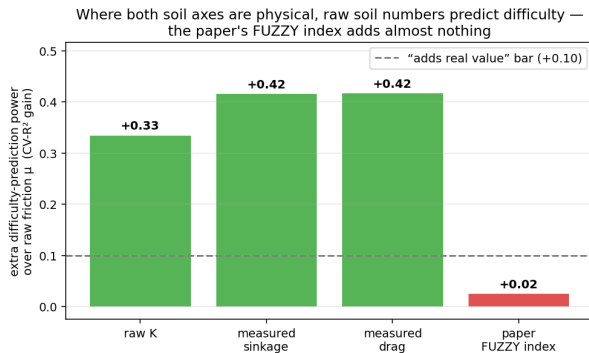
The support axis is now real:

- Soft soil ($K=1$): feet sink **~49 mm**.
- Firm soil ($K=8$): sink **~23 mm**; drag drops **~45 → 20 N**.
- Confirmed by a force-off control (no forces $\Rightarrow$ no effect).

Same robot, same gait — only the soil firmness changes. You can see it.

Set C: the key result

Now **both** soil dials are physical. Does the fuzzy index finally add value?



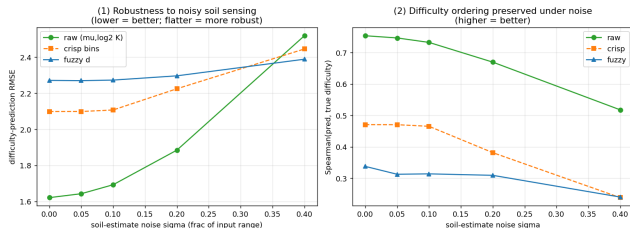
- Raw soil numbers **strongly predict** difficulty (+0.33 to +0.42).
- The paper's **fuzzy index** adds **+0.025** — essentially nothing (statistically detectable, $p=0.001$, but practically zero).

Sharp finding

Even where firmness **is** physically active, the fuzzy mapping **fails to use it**. The problem is the **fuzzy model's structure**, **not** the simulator.

Giving fuzzy its best shot: noisy soil sensing

Fuzzy logic's real strength is **tolerating imprecise inputs**. A real robot only *estimates* the soil. So we added sensing noise.



- Fuzzy **is** noise-stable (flat line) and edges raw on error-size *only* at extreme noise.
- But it “wins” by being **stable, not accurate** — even there, raw still **orders soils better**. Flatness $\neq$ usefulness.
- **Decisions (②)**: a fuzzy-driven curriculum **camps on easy soil** and never advances.

Stability of an *uninformative* signal — it doesn't swing because it barely says anything.

Set D: can we *fix* the broken fuzzy rules?

Set C blamed the **rule structure** (it ignores firmness). So we redesigned it — **only** the rule table, so support now matters at *every* traction level.

Descriptor	Uses firmness?
paper fuzzy index	+0.025 (blind)
redesigned fuzzy	+0.416 (sees it)
raw soil numbers	+0.33 to +0.42

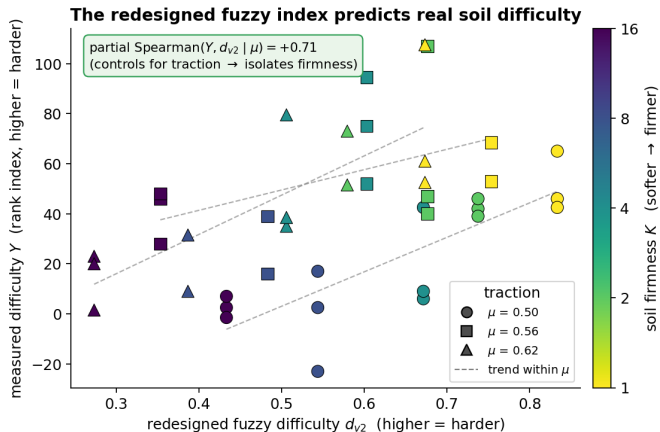
Cross-validated R^2 gain over friction alone (higher = uses firmness more). The redesign lands **inside** the raw numbers' range — it *matches* them, not beats.

The fix works — as a descriptor:

- Firmness signal jumps **0.025** → **0.416** ($\sim 16\times$).
- The original failure was **structural and fixable**, not the simulator.

Honest: the fix is informed by the data's firm=easier trend — a **corrected** descriptor, not a vindication of the paper's rules. Raw soil numbers stay the ceiling; the fix only **matches** them, never exceeds.

Does the redesigned fuzzy index actually work? (descriptor check)



Within the measured sweep, the redesigned fuzzy index **orders real difficulty** and **separates firmness**: within each traction μ , a higher fuzzy score = harder soil (partial Spearman +0.71).

Descriptor validation only — d_{v2} was tuned on this regime; it aids interpretation, not a claim that fuzzy beats raw for training.

Set D: but does the fix help the robot *train*?

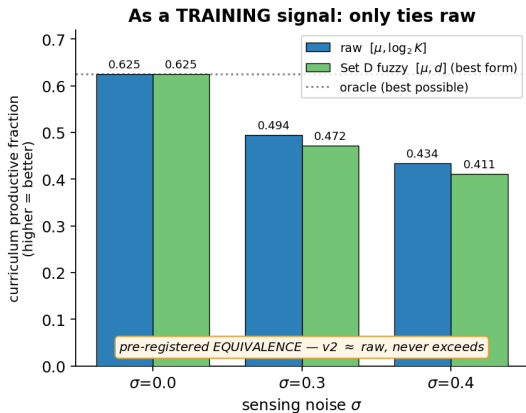
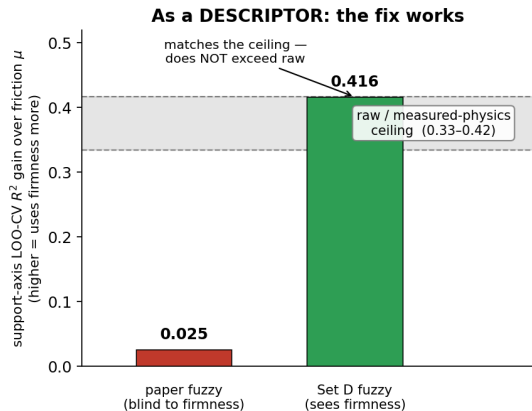
The catch: a better *description* of soil is not the same as a better *training signal*. We pace the curriculum (when to move to harder soil) with the fixed index — cheap gated proxy, vetted with an independent model.

- As a single score: **fails** — it trails the raw numbers and thrashes.
- In its **best form** $[\mu, \text{fuzzy}]$: **fully fixes** the camping and **matches** the raw numbers' accuracy...
- ...but only **ties** them — it carries the **same information** as raw $[\mu, \log_2 K]$, never beats it (a pre-registered **equivalence** result: a tie, not a win).

Decision

GPU training run not justified (~ 24 h saved). A better *description* of the soil does **not** become a better *training signal* than the raw numbers.

Set D in one picture: a descriptor win, not a training win



Fixing the fuzzy **rule table** makes the index physically meaningful as a **descriptor** (left: 0.025 $\rightarrow$ 0.416, reaching the raw/physics ceiling) — but as a **training** signal it only **ties** raw soil numbers (right: equivalent across noise). **Raw stays the practical ceiling.**

Bottom line

Across rigid **and** deformable soil, the paper's fuzzy soil index shows **no demonstrable value** over the raw soil numbers — as a label, a curriculum, a descriptor, or under noisy sensing. And when we **fix** it, it becomes a good descriptor but **still only ties** raw for training.

- **Replicates:** the friction-curriculum **survival** benefit; DFH genuinely restores the firmness axis.
- **Does not add value:** the paper's mapping collapses two soil factors into a coarse score and **discards the useful information**.
- The failure is **structural** (Set C), and **fixable** (Set D) — but the fix buys **interpretability, not performance**: raw is the ceiling fuzzy can reach, not exceed.

Scope & what would change the answer

Honest scope:

- This is about **this paper's specific fuzzy mapping** — *not* all 2-D or learned terrain descriptors.
- DFH is an **uncalibrated** deformable-soil surrogate; one frozen walker.
- Not a disproof of the paper on **real** soil — a non-replication *in simulation*.

What could make “fuzzy matter”:

- ~~Redesign the rules to use the firmness axis~~ — **done (Set D)**: it works as a **descriptor** but still only **ties** raw for training.
- A genuinely **new** claim with **noise-stability as the goal**, or a setting where the raw soil numbers **aren't available** (e.g. a real robot with only a coarse visual soil label, no precise sensors).
- Train a soil-robust walker (removes the single-policy caveat).

A fuzzy score is only useful if it beats the numbers it's made from.

Here it doesn't — even after we fixed it, the best it does is **tie**.

*Fuzzy soil difficulty is a strong, interpretable **description** of terrain hardness, but it does not improve curriculum **performance** over the raw soil parameters for Hunter locomotion.*

Code & full write-ups: `docs/experiments/fuzzy_soil_UNIFIED_conclusion.md`; models on Hugging Face `anhrisn/hunter-dfh-locomotion`.